

Elementary & Intermediate Algebra



Solving linear equations

1 Solve for x : $4x - 9 = 23$.

2 Solve for x : $\frac{3x - 2}{5} = 8$.

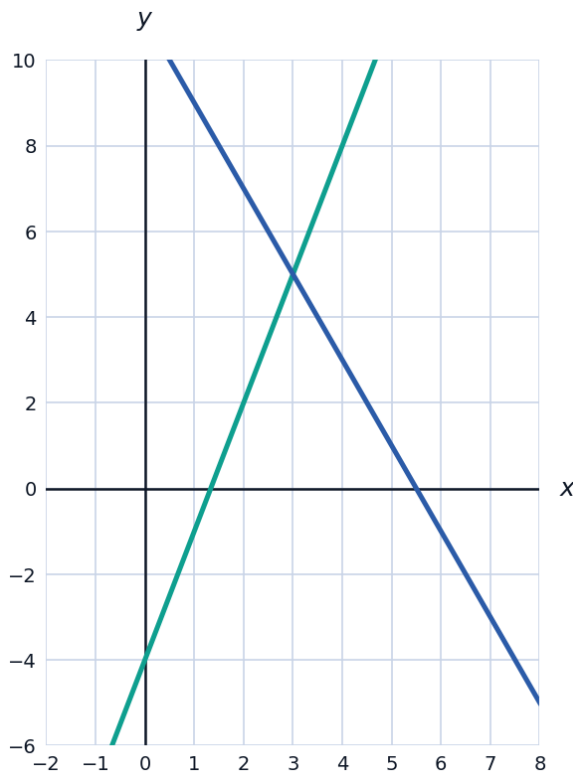
Solving and graphing inequalities

3 Solve $5x + 3 < 28$, giving your answer as an inequality.

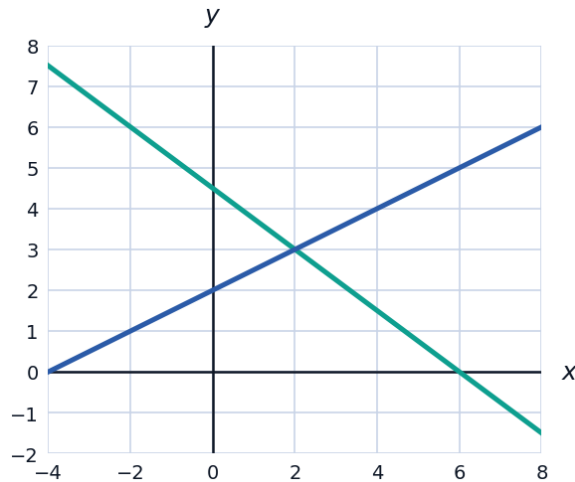
4 Solve $-4x + 7 \leq -9$, being careful with the direction of the inequality.

Systems of linear equations

5 Solve the system: $y = 3x - 4$ and $2x + y = 11$.



- 6 Solve the system: $3x + 4y = 18$ and $x - 2y = -4$, using elimination.



Factoring and solving quadratic equations

- 7 Solve $x^2 - x - 12 = 0$ by factoring.
- 8 Solve $2x^2 - 5x - 3 = 0$ by factoring.

Simplifying rational and radical expressions

- 9 Simplify $\frac{x^2 - 9}{x + 3}$.
- 10 Simplify $\sqrt{48}$.

Working with exponents

- 11 Simplify $\frac{x^7}{x^3}$.
- 12 Simplify $(3x^2y)^3$.

Absolute value equations

- 13 Solve $|3x + 2| = 11$.
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Word problems leading to linear or quadratic equations

- 14 The length of a rectangle is 3 more than twice its width, and its area is 65. Find the width and length.



- 15 A number increased by 8 is 3 times the original number decreased by 4. Find the number.
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Combining algebraic techniques in a multi-step problem

- 16 This question has two parts. Two consecutive positive integers have a product of 132. (a) Set up and solve a quadratic equation to find the two integers. (b) Verify your answer by checking the product.
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Solving quadratics using the quadratic formula

- 17 Solve $x^2 + 3x - 2 = 0$ using the quadratic formula, giving exact answers.
- 18 Use the discriminant to determine the number of real solutions of $4x^2 - 12x + 9 = 0$, without solving.
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Factoring by grouping and special patterns

- 19 Factor $x^3 + 2x^2 - 9x - 18$ by grouping.
- 20 Factor $4x^2 - 25$ completely.
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Rational equations and extraneous solutions

- 21 Solve $\frac{x}{x-2} = \frac{3}{x-2} + 1$, checking for extraneous solutions.
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Multi-step algebraic word problems

- 22** This question has two parts. A boat travels 60 miles downstream in 3 hours and returns upstream in 5 hours. (a) Let b be the boat's speed in still water and c be the current's speed; write two equations based on the trip times. (b) Solve the system to find b and c .